Workshop: One-Way ANOVA and Kruskal-Wallis

your name here

This week, we continue working with the Totally Serious Pop Culture Research Institute™ on research questions involving multiple groups.

Let's call in the packages and data we need,

```
library(tidyverse)
library(ssstats)
binge_watch  <-  read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M2-binge_watch.csv")
```

```
Error:
! cannot open the connection
```

```
franchise_events <- read_csv("https://raw.githubusercontent.com/samanthaseals/
STA4173Fa26/refs/heads/main/workshops/data/M2-franchise_events.csv")
```

```
Error:
! cannot open the connection
```

```
training_performance        <-        read_csv("https://raw.githubusercontent.
com/samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-training_
performance.csv")
```

```
Error:
! cannot open the connection
```

```
villain_menace <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M2-villain_menace.csv")
```

```
Error:
! cannot open the connection
```

Study 1: Villain Archetype Power Levels

The Totally Serious Pop Culture Research Institute conducted an experiment to determine whether different villain archetypes produce different levels of perceived menace. Researchers recruited 90 participants and randomly assigned 30 participants to each of three conditions: The Dark Lord archetype, The Misunderstood Anti-Hero archetype, and The Comically Incompetent archetype.

Participants watched a 2-minute clip featuring a villain of their assigned archetype and then rated the villain's menace on a scale from 0 to 100, with higher scores indicating greater perceived menace. The researchers want to determine whether average menace ratings differ across the three villain archetypes.

Dataset name: `villian_menace`

Variables:

- `archetype`
- `menace_rating`

1a. Find the summary statistics to describe this data.

```
villain_menace |> group_by(archetype) |> mean_median(menace_rating)
```

```
Error:
! object 'villain_menace' not found
```

1b. Construct the appropriate q-q plot for this dataset and research question.

```
villain_menace |> anova_check(outcome = menace_rating,
                              grouping = archetype)
```

```
Error:
! object 'villain_menace' not found
```

1c. Based on your q-q plot in 1b, do we meet the necessary normality assumption? Why or why not?

Your answer here.

1d. Based on your answers in 1b and 1c, would you test the means or the medians here? Why?

Your answer here.

1e. Construct the appropriate hypothesis test to determine if average menace ratings differ across the three villain archetypes.

```
villain_menace |> anovas(outcome = menace_rating,
                          grouping = archetype)
```

```
Error:
! object 'villain_menace' not found
```

```
villain_menace |> kruskal_wallis(outcome = menace_rating,
                          grouping = archetype)
```

```
Error:
! object 'villain_menace' not found
```

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

1f. Perform an appropriate posthoc test to identify which villain archetypes differ significantly in menace ratings. Let's treat this as a confirmatory study.

```
villain_menace |> posthoc(outcome = menace_rating,
                           grouping = archetype)
```

```
Error:
! object 'villain_menace' not found
```

```
villain_menace |> posthoc(outcome = menace_rating,
                           grouping = archetype,
                           nonparametric = TRUE)
```

```
Error:
! object 'villain_menace' not found
```

1g. Which groups, if any, are significantly different from one another?

Your answer here.

Study 2: Superhero Training Program Performance

The Totally Serious Pop Culture Research Institute tracks the effectiveness of three different superhero training programs. They recruited 75 participants and randomly assigned 25 participants to each program: The Classical Hero Academy, The Gritty Realism Program, and The Anime-Inspired Intense Program.

After 12 weeks of training, each participant completed a standardized performance assessment scored from 0 to 100, with higher scores indicating better overall performance. The researchers want to determine whether average performance scores differ across the three training programs.

Dataset name: `training_performance`

Variables:

- `program`
- `performance_score`

2a. Find the summary statistics to describe this data.

```
training_performance |> group_by(program) |> mean_median(performance_score)
```

```
Error:
! object 'training_performance' not found
```

2b. Construct the appropriate q-q plot for this dataset and research question.

```
training_performance |> anova_check(outcome = performance_score,
                                    grouping = program)
```

```
Error:
! object 'training_performance' not found
```

2c. Based on your q-q plot in 2b, explain why Dr. Seals believes we meet the ANOVA assumptions.

Your answer here.

2d. Construct the appropriate hypothesis test (ANOVA) to determine if average performance scores differ across the three training programs.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

2e. Perform an appropriate posthoc test to identify which training programs differ significantly in performance. Let's treat this as an exploratory study.

2f. Which groups, if any, are significantly different from one another?

Your answer here.

Study 3: Streaming Service Binge-Watch Behavior

The Totally Serious Pop Culture Research Institute investigated binge-watching habits across subscribers to three different streaming platforms. They recruited 80 participants who actively use one of three services: StreamFlix, PrimateMax, or CinemaPlus.

Over a two-week period, researchers counted the number of episodes each participant watched of their favorite show. The researchers want to determine whether the median number of episodes watched differs across the three streaming platforms.

Dataset name: `binge_watch`

Variables:

- `platform`
- `episodes_watched`

3a. Find the summary statistics to describe this data.

```
binge_watch |> group_by(platform) |> mean_median(episodes_watched)
```

```
Error:
! object 'binge_watch' not found
```

3b. Construct the appropriate q-q plot for this dataset and research question.

```
binge_watch |> anova_check(outcome = episodes_watched,
                           grouping = platform)
```

```
Error:
! object 'binge_watch' not found
```

3c. Based on your q-q plot in 3b, do we meet the necessary normality assumption? Why or why not?

Your answer here.

3d. Based on your answers in 3b and 3c, would you test the means or the medians here? Why?

Your answer here.

3e. Construct the appropriate hypothesis test to determine if the number of episodes watched differs across the three streaming platforms.

```
binge_watch |> anovas(outcome = episodes_watched,
                      grouping = platform)
```

```
Error:
! object 'binge_watch' not found
```

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

3f. Explain why it is not appropriate to perform posthoc testing in this situation.

Your answer here.

Study 4: Fan Engagement Across Franchises

The Totally Serious Pop Culture Research Institute surveyed fan engagement in four major entertainment franchises. They recruited 100 fans (25 per franchise) from four franchises: The Galaxian Wars, Marvel Legends, The Wizard Chronicles, and Dark Knight Studios.

Researchers counted how many official franchise events each fan attended in the past year. The researchers want to determine whether the median number of events attended differs across the four franchises.

Dataset name: `franchise_events`

Variables:

- `franchise`
- `events_attended`

4a. Find the summary statistics to describe this data.

```
franchise_events |> group_by(franchise) |> mean_median(events_attended)
```

```
Error:
! object 'franchise_events' not found
```

4b. Construct the appropriate q-q plot for this dataset and research question.

```
franchise_events |> anova_check(outcome = events_attended,
                                grouping = franchise)
```

```
Error:
! object 'franchise_events' not found
```

4c. Based on your q-q plot in 4b, do we meet the necessary normality assumption? Why or why not?

Your answer here.

4d. Based on your answers in 4b and 4c, would you test the means or the medians here? Why?

Your answer here.

4e. Construct the appropriate hypothesis test to determine if the median number of events attended differs across the four franchises.

```
franchise_events |> anovas(outcome = events_attended,
                           grouping = franchise)
```

```
Error:
! object 'franchise_events' not found
```

```
franchise_events |> kruskal_wallis(outcome = events_attended,
                                    grouping = franchise)
```

```
Error:
! object 'franchise_events' not found
```

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

4f. Perform an appropriate posthoc test to identify which franchises differ significantly in event attendance. Let's treat this as an exploratory study.

4g. Which groups, if any, are significantly different from one another?

Your answer here.